

Xist RNA is a potent suppressor of hematologic cancer in mice.

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X chromosome aneuploidies have long been associated with human cancers, but causality has not been established. In mammals, X chromosome inactivation (XCI) is triggered by Xist RNA to equalize gene expression between the sexes. Here we delete Xist in the blood compartment of mice and demonstrate that mutant females develop a highly aggressive myeloproliferative neoplasm and myelodysplastic syndrome (mixed MPN/MDS) with 100% penetrance. Significant disease components include primary myelofibrosis, leukemia, histiocytic sarcoma, and vasculitis. Xist-deficient hematopoietic stem cells (HSCs) show aberrant maturation and age-dependent loss. Reconstitution experiments indicate that MPN/MDS and myelofibrosis are of hematopoietic rather than stromal origin. We propose that Xist loss results in X reactivation and consequent genome-wide changes that lead to cancer, thereby causally linking the X chromosome to cancer in mice. Thus, Xist RNA not only is required to maintain XCI but also suppresses cancer in vivo.

Xist exerts gene-specific silencing during XCI maintenance and impacts lineage-specific cell differentiation and proliferation during hematopoiesis.

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X chromosome inactivation (XCI) is a dosage compensation phenomenon that occurs in females. Initiation of XCI depends on Xist RNA, which triggers silencing of one of the two X chromosomes, except for XCI escape genes that continue to be biallelically expressed. In the soma XCI is stably maintained with continuous Xist expression. How Xist impacts XCI maintenance remains an open question. Here we conditionally delete Xist in hematopoietic system of mice and report differentiation and cell cycle defects in female hematopoietic stem and progenitor cells (HSPCs). By utilizing female HSPCs and mouse embryonic fibroblasts, we find that X-linked genes show variable tolerance to Xist loss. Specifically, XCI escape genes exhibit preferential transcriptional upregulation, which associates with low H3K27me3 occupancy and high chromatin accessibility that accommodates preexisting binding of transcription factors such as Yin Yang 1 (YY1) at the basal state. We conclude that Xist is necessary for gene-specific silencing during XCI maintenance and impacts lineage-specific cell differentiation and proliferation during hematopoiesis.

Evolution from XIST-independent to XIST-controlled X-chromosome inactivation: epigenetic modifications in distantly related mammals.

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X chromosome inactivation (XCI) is the transcriptional silencing of one X in female mammals, balancing expression of X genes between females (XX) and males (XY). In placental mammals non-coding XIST RNA triggers silencing of one X (Xi) and recruits a characteristic suite of epigenetic modifications, including the histone mark H3K27me3. In marsupials, where XIST is missing, H3K27me3 association seems to have different degrees of stability, depending on cell-types and species. However, the complete suite of histone marks associated with the Xi and their stability throughout cell cycle remain a mystery, as does the evolution of an ancient mammal XCI system. Our extensive immunofluorescence analysis (using

antibodies against specific histone modifications) in nuclei of mammals distantly related to human and mouse, revealed a general absence from the mammalian Xi territory of transcription machinery and histone modifications associated with active chromatin. Specific repressive modifications associated with XCI in human and mouse were also observed in elephant (a distantly related placental mammal), as was accumulation of XIST RNA. However, in two marsupial species the Xi either lacked these modifications (H4K20me1), or they were restricted to specific windows of the cell cycle (H3K27me3, H3K9me2). Surprisingly, the marsupial Xi was stably enriched for modifications associated with constitutive heterochromatin in all eukaryotes (H4K20me3, H3K9me3). We propose that marsupial XCI is comparable to a system that evolved in the common therian (marsupial and placental) ancestor. Silent chromatin of the early inactive X was exapted from neighbouring constitutive heterochromatin and, in early placental evolution, was augmented by the rise of XIST and the stable recruitment of specific histone modifications now classically associated with XCI.

RNF12 activates Xist and is essential for X chromosome inactivation.

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In somatic cells of female placental mammals, one of the two X chromosomes is transcriptionally silenced to accomplish an equal dose of X-encoded gene products in males and females. Initiation of random X chromosome inactivation (XCI) is thought to be regulated by X-encoded activators and autosomally encoded suppressors controlling Xist. Spreading of Xist RNA leads to silencing of the X chromosome in cis. Here, we demonstrate that the dose dependent X-encoded XCI activator RNF12/RLIM acts in trans and activates Xist. We did not find evidence for RNF12-mediated regulation of XCI through Tsix or the Xist intron 1 region, which are both known to be involved in inhibition of Xist. In addition, we found that Xist intron 1, which contains a pluripotency factor binding site, is not required for suppression of Xist in undifferentiated ES cells. Analysis of female *Rnf12*^{-/-} knockout ES cells showed that RNF12 is essential for initiation of XCI and is mainly involved in the regulation of Xist. We conclude that RNF12 is an indispensable factor in up-regulation of Xist transcription, thereby leading to initiation of random XCI.

Downregulation of lncRNA XIST may promote Th17 differentiation through KDM6A-TSAd pathway in neuromyelitis optica spectrum disorders.

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Neuromyelitis optica spectrum disorder (NMOSD) is an autoimmune disease with significant female preponderance. X inactive specific transcript (XIST) is a long non-coding RNA (lncRNA) and a key regulator of X-chromosome inactivation which is related to the sex-bias of autoimmunity. And Th17 cell proportion was significantly elevated in NMOSD according to our previous study. This study aimed to explore the expression levels of lncRNA XIST-KDM6A-TSAd pathway in lymphocytes of female NMOSD patients, and investigate its possible relationship with pathogenesis of NMOSD. The study enrolled 30 acute-phase untreated female NMOSD patients and 30 age-matched female healthy controls, their lymphocytes were collected for experiments. Microarray as well as validation experiments showed lncRNA XIST was significantly downregulated in the NMOSD group. And the levels of lysine demethylase 6A (KDM6A) decreased in NMOSD and showed significant positive correlation with XIST. The levels of T cell-specific adapter (TSAd) mRNA and protein levels were significantly lower in NMOSD. And Chromatin

immunoprecipitation assay demonstrated that NMOSD had more H3K27me3 modification than control at TSAd promoter region. The present study introduced a potential pathway that following lncRNA XIST downregulation, which process may promote Th17 differentiation in NMOSD. These findings shed new light on the immune regulation mechanism about lncRNA XIST and related epigenetic features, which may contribute to develop female-specific treatment plans.

Time-resolved structured illumination microscopy reveals key principles of Xist RNA spreading.

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X-inactive specific transcript (Xist) RNA directs the process of X chromosome inactivation in mammals by spreading in cis along the chromosome from which it is transcribed and recruiting chromatin modifiers to silence gene transcription. To elucidate mechanisms of Xist RNA cis-confinement, we established a sequential dual-color labeling, super-resolution imaging approach to trace individual Xist RNA molecules over time, which enabled us to define fundamental parameters of spreading. We demonstrate a feedback mechanism linking Xist RNA synthesis and degradation and an unexpected physical coupling between preceding and newly synthesized Xist RNA molecules. Additionally, we find that the protein SPEN, a key factor for Xist-mediated gene silencing, has a distinct function in Xist RNA localization, stability, and coupling behaviors. Our results provide insights toward understanding the distinct dynamic properties of Xist RNA.

Developmental Xist induction is mediated by enhanced splicing.

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X-inactive-specific transcript (Xist) is a long noncoding RNA (lncRNA) essential for inactivating one of the two X chromosomes in mammalian females. Random X chromosome inactivation is mediated by Xist RNA expressed from the inactive X chromosome. We found that Xist RNA is unspliced in naïve embryonic stem (ES) cells. Upon differentiation, Xist splicing becomes efficient across all exons independent of transcription, suggesting interdependent or coordinated removal of Xist introns. In female cells with mutated polypyrimidine tract binding protein 1 (Ptbp1), differentiation fails to substantially upregulate mature Xist RNA because of a defect in Xist splicing. We further found both Xist129 and XistCAS RNA are unspliced in *Mus musculus* 129SvJ/*Mus castaneus* (CAS) hybrid female ES cells. Upon differentiation, Xist129 exhibits a higher splicing efficiency than XistCAS, likely contributing to preferential inhibition of the X129 chromosome. Single cell analysis shows that the allelic choice of Xist splicing is linked to the inactive X chromosome. We conclude post-transcriptional control of Xist RNA splicing is an essential regulatory step of Xist induction. Our studies shed light on the developmental roles of splicing for nuclear-retained Xist lncRNA and suggest inefficient Xist splicing is an additional fail-safe mechanism to prevent Xist activity in ES cells.

X chromosome inactivation and the Xist gene.

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Recent years have seen rapid progress towards understanding the molecular mechanisms involved in X chromosome inactivation (X inactivation). This progress has largely revolved around the discovery of the X inactive specific transcript (Xist) gene, which is known now to represent the master switch locus regulating X inactivation. In adult cells Xist is transcribed exclusively from the inactive X chromosome. The transcript has no apparent protein-coding potential and is retained in the nucleus in close association with the domain occupied by the inactive X chromosome. It is thus thought to represent a functional RNA molecule which acts as the primary signal responsible for the propagation of X inactivation. Developmental regulation of Xist correlates with the developmental timing of X inactivation. Recent results have demonstrated that Xist is both necessary and sufficient for X inactivation. Goals for the future are to understand the mechanism of Xist regulation which underlies the establishment of appropriate X inactivation patterns and to determine how Xist RNA participates in the process of propagating inactivation in cis.

A scaffold for X chromosome inactivation.

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X chromosome inactivation (XCI), the silencing of one of the two X chromosomes in XX female cells, equalises the dosage of X-linked genes relative to XY males. The process is mediated by the non-coding RNA X inactive specific transcript (Xist) that binds in cis and propagates along the inactive X chromosome elect, triggering chromosome-wide silencing. The mechanisms by which Xist RNA binds and spreads along the chromosome, and initiates Xist-mediated chromosome silencing remain poorly understood. Accumulating evidence suggests that chromosome and nuclear organisation are important in both processes. Notably, recent studies have identified specific factors, previously shown to be components of the nuclear matrix or scaffold, to play a role both in Xist RNA-binding and in Xist-mediated silencing. In this review we provide a perspective on these studies in the context of previous work on chromosome/nuclear architecture in XCI.

The B-side of Xist.

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Female mammals express the long noncoding X inactivation-specific transcript (Xist) RNA to initiate X chromosome inactivation (XCI) that eventually results in the formation of the Barr body. Xist encompasses half a dozen repeated sequence stretches containing motifs for RNA-binding proteins that recruit effector complexes with functions for silencing genes and establishing a repressive chromatin configuration. Functional characterization of these effector proteins unveils the cooperation of a number of pathways to repress genes on the inactive X chromosome. Mechanistic insights can be extended to other noncoding RNAs with similar structure and open avenues for the design of new therapies to switch off gene expression. Here we review recent advances in the understanding of Xist and on this basis try to synthesize a model for the initiation of XCI.